# Topic quiz: Circular motion

IBDP Physics SL · Grade 11 · A.2 Circular motion · 10 multiple-choice questions

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Name: Class: Date:

**How to use.** Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use g = 9.81 m s$^{-2}$ and show any working next to the question.
**Check yourself online:** the same quiz with instant feedback and explanations — https://school.taneski.com/physics/dp/quiz-circular-motion/

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
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### Question 1

An object moves in a horizontal circle at constant speed. Which statement is correct?

(A) Its velocity is constant, so there is no resultant force
(B) Its velocity changes, and its acceleration points towards the centre
(C) Its acceleration points along its direction of motion
(D) Its acceleration points away from the centre

### Question 2

A child sits 2.0 m from the centre of a playground carousel that makes one turn every 4.0 s. What is the child's speed?

(A) 0.50 m s$^{-1}$
(B) 1.6 m s$^{-1}$
(C) 3.1 m s$^{-1}$
(D) 4.9 m s$^{-1}$

### Question 3

A 1200 kg car drives round a bend of radius 50 m at 15 m s$^{-1}$. What is the centripetal force on the car?

(A) 360 N
(B) 2700 N
(C) 5400 N
(D) 11 800 N

### Question 4

A car goes round a flat (unbanked) bend at constant speed. What provides the centripetal force?

(A) Friction between the tyres and the road, acting towards the centre
(B) A centrifugal force acting outwards
(C) The driving force of the engine
(D) The normal force of the road

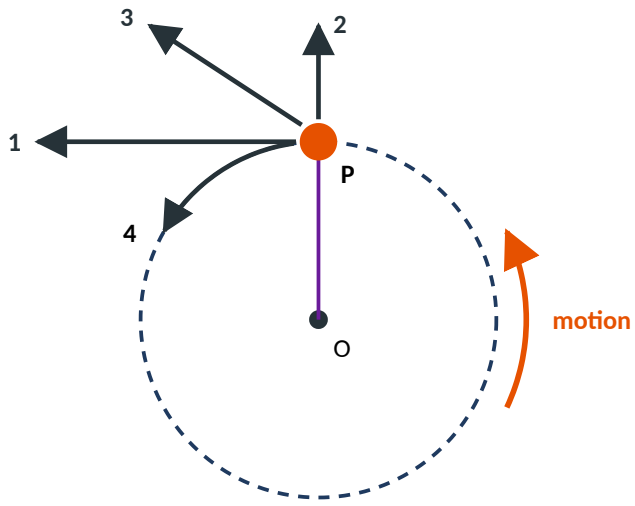
### Question 5

On a wet road $\mu_s = 0.40$. What is the maximum speed at which a car can take a flat bend of radius 50 m without skidding?

(A) 3.9 m $s^{-1}$
(B) 14.0 m $s^{-1}$
(C) 22.1 m $s^{-1}$
(D) 196 m $s^{-1}$

### Question 6

A ball on a string moves in a horizontal circle (seen from above). The string breaks when the ball is at P. Which path does the ball follow?




view from above

(A) Path 1
(B) Path 2
(C) Path 3
(D) Path 4

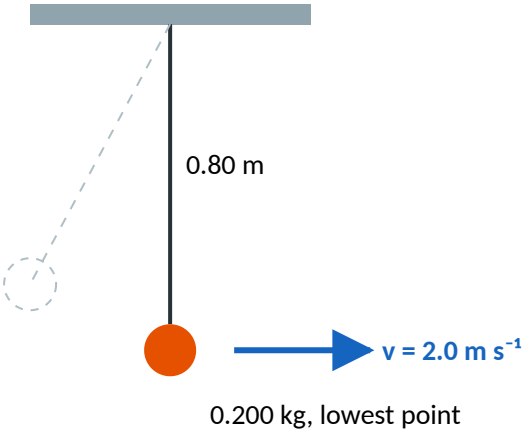
### Question 7

The International Space Station orbits at radius $6.77 \times 10^6$ m from Earth's centre with a speed of $7.67 \times 10^3$ m $s^{-1}$. What is the centripetal acceleration of an astronaut on board?

(A) 0 m $s^{-2}$ — the astronaut is weightless
(B) $1.13 \times 10^{-3}$ m $s^{-2}$
(C) 8.69 m $s^{-2}$
(D) 9.81 m $s^{-2}$

### Question 8

A 0.200 kg pendulum bob on a 0.80 m string passes through its lowest point at 2.0 m $s^{-1}$. What is the tension in the string there?




(A) 0.96 N
(B) 1.00 N
(C) 1.96 N
(D) 2.96 N

### Question 9

The angular speed of a carousel is doubled. What happens to the centripetal acceleration of a child sitting at the same place?

(A) It halves
(B) It stays the same
(C) It doubles
(D) It becomes four times as large

### Question 10

Two children sit on the same carousel: A is 1.0 m from the centre and B is 2.0 m from the centre. Which statement is correct?

(A) They have the same speed and the same acceleration
(B) B has twice the speed of A and twice the acceleration
(C) B has twice the speed of A and the same acceleration
(D) B has twice the speed of A and half the acceleration

End of quiz · Score: _____ / 10